

PS3. Q1.1

$$\frac{dx}{dt} = x(1+x-2y)$$

$$\frac{dy}{dt} = y(x-1)$$

EQUILIBRIUM POINTS

$$x(1+x-2y) = 0$$

$$y(x-1) = 0$$

$$y(x-1) = 0 \Leftrightarrow \begin{matrix} (1) \\ y=0 \end{matrix} \vee \begin{matrix} (2) \\ x=1 \end{matrix}$$

$$\bullet (1) \quad x(1+x) = 0 \Leftrightarrow x=0 \vee x=-1$$

$$\bullet (2) \quad (2-2y) = 0 \Leftrightarrow y=1$$

$$\begin{pmatrix} x_{e1} \\ y_{e1} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}; \quad \begin{pmatrix} x_{e2} \\ y_{e2} \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}; \quad \begin{pmatrix} x_{e3} \\ y_{e3} \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

JACOBIAN

$$J = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix} = \begin{bmatrix} 1+2x-2y & -2x \\ y & x-1 \end{bmatrix}$$

LINEARISED SYSTEM, E.P. CLASSIFICATION (check the Δ - τ diagram)

$$J(0,0) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \Rightarrow \begin{matrix} \tau = 0 \\ \Delta = -1 \end{matrix} \Rightarrow \text{saddle point}$$

$$J(-1,0) = \begin{pmatrix} -1 & 2 \\ 0 & -2 \end{pmatrix} \Rightarrow \begin{matrix} \tau = -3 \\ \Delta = 2 \\ \tau^2 - 4\Delta = 9 - 8 = 1 \end{matrix} \Rightarrow \text{attractor node}$$

$$J(1,1) = \begin{pmatrix} 1 & -2 \\ 1 & 0 \end{pmatrix} \Rightarrow \begin{matrix} \tau = 1 \\ \Delta = 2 \\ \tau^2 - 4\Delta = 1 - 8 = -7 \end{matrix} \Rightarrow \text{unstable spiral}$$

PS3. Q1.2

$$\frac{dx}{dt} = 4 - 4x^2 - y^2$$

$$\frac{dy}{dt} = 3xy$$

EQUILIBRIUM POINTS

$$4 - 4x^2 - y^2 = 0$$

$$3xy = 0$$

$$3xy = 0 \Leftrightarrow \begin{matrix} (1) \\ x=0 \end{matrix} \vee \begin{matrix} (2) \\ y=0 \end{matrix}$$

$$(1) \quad 4 - y^2 = 0 \Leftrightarrow y = \pm \sqrt{2}$$

$$(2) \quad 4 - 4x^2 = 0 \Leftrightarrow x = \pm 1$$

$$\begin{pmatrix} x_{e1} \\ y_{e1} \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}; \begin{pmatrix} x_{e2} \\ y_{e2} \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \end{pmatrix}; \begin{pmatrix} x_{e3} \\ y_{e3} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}; \begin{pmatrix} x_{e4} \\ y_{e4} \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

JACOBIAN

$$J = \begin{bmatrix} -8x & -2y \\ 3y & 3x \end{bmatrix}$$

LINEARISED SYSTEM, E.P. CLASSIFICATION (check the Δ - τ diagram)

$$J(0,2) = \begin{bmatrix} 0 & -4 \\ 6 & 0 \end{bmatrix} \Rightarrow \begin{matrix} \tau = 0 \\ \Delta = 24 \end{matrix} \Rightarrow \text{centre}$$

$$J(0,-2) = \begin{bmatrix} 0 & 4 \\ -6 & 0 \end{bmatrix} \Rightarrow \begin{matrix} \tau = 0 \\ \Delta = 24 \end{matrix} \Rightarrow \text{centre}$$

$$J(1,0) = \begin{bmatrix} -8 & 0 \\ 0 & 3 \end{bmatrix} \Rightarrow \begin{matrix} \tau = -5 \\ \Delta = -24 \end{matrix} \Rightarrow \text{saddle point}$$

$$J(-1,0) = \begin{bmatrix} 8 & 0 \\ 0 & -3 \end{bmatrix} \Rightarrow \begin{matrix} \tau = 5 \\ \Delta = -24 \end{matrix} \Rightarrow \text{saddle point}$$

PS3, Q2.1

$$\frac{dx}{dt} = -x - y$$

$$\frac{dy}{dt} = -x + y + 2xy$$

EQUILIBRIUM POINTS

$$-x - y = 0$$

$$-x + y + 2xy = 0$$

$$-x - y = 0 \Rightarrow y = -x$$

$$-x - x - 2x^2 = 0 \Rightarrow -2x(1+x) = 0 \Rightarrow x = 0 \vee x = -1$$

$$\underline{p}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \underline{p}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

JACOBIAN

$$J = \begin{bmatrix} -1 & -1 \\ -1+2y & 1+2x \end{bmatrix}$$

LINEARISED SYSTEM, E.P. CLASSIFICATION (check the Δ - τ diagram)

$$J(0,0) = \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} \Rightarrow \tau = 0, \Delta = -2 \Rightarrow \text{saddle point}$$

$$J(-1,1) = \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix} \Rightarrow \tau = -2, \Delta = 2, \tau^2 - 4\Delta = 4 - 8 = -4 \Rightarrow \text{stable spiral}$$

ISOCINES

$$K = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{-x+y+2xy}{-x-y}$$

$$K=0 \Leftrightarrow -x+y+2xy=0 \Leftrightarrow y = \frac{x}{1+2x}$$

ARROWS ARE HORIZONTAL
(EITHER $\rightarrow$ OR $\leftarrow$)

$$\frac{dx}{dt} > 0 \text{ if } -x-y > 0 \Leftrightarrow y < -x$$

BELOW THE
BISECTOR OF
THE 2nd AND 4th
QUADRANTS ARROWS
GO RIGHT,
OTHERWISE LEFT.

• $K = \infty \Rightarrow -x - y = 0 \Rightarrow y = -x$

↑
ARROWS ARE VERTICAL
(EITHER ↑ OR ↓)

$\frac{dy}{dt} > 0$ if $-x - x - 2x^2 > 0 \Rightarrow -2x(1+x) > 0$

Hence, if $-1 < x < 0$ ARROWS POINT UP
OTHERWISE DOWN

• $K = -1$

↑
ARROWS ARE INCLINED AT
 -45° (EITHER ↙ OR ↘)

$-x - y = x - y - 2xy \Rightarrow 2x(y-1) = 0$

$x = 0$

$y = 1$

• $x = 0 \Rightarrow \frac{dx}{dt} = -y > 0$ if $y < 0$ (DOWN-RIGHT)

• $y = 1 \Rightarrow \frac{dx}{dt} = -x - 1 > 0$ if $x < -1$ (DOWN-LEFT)

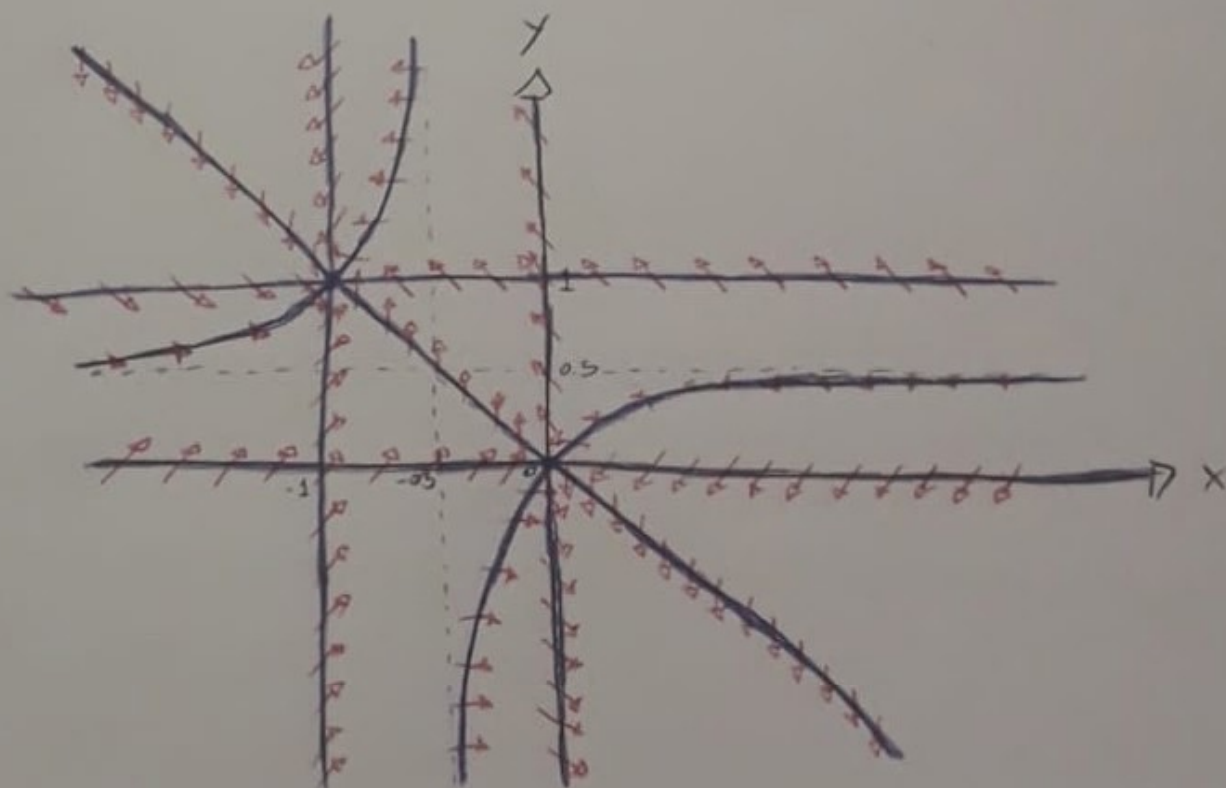
• $K = 1$

↑
ARROWS ARE INCLINED AT
 45° (EITHER ↗ OR ↘)

$-x - y = -x + y + 2xy \Rightarrow 2y(x+1) = 0 \Rightarrow x = -1$

• $x = -1 \Rightarrow \frac{dx}{dt} = 1 - y > 0$ if $y < 1$ (UP-LEFT)

• $y = 0 \Rightarrow \frac{dx}{dt} = -x > 0$ if $x < 0$ (UP-LEFT)



83 Q2.2

$$\frac{dx}{dt} = x - y - 2xy$$

$$\frac{dy}{dt} = 4x + y$$

EQUILIBRIUM POINTS

$$\begin{aligned} x - y - 2xy &= 0 \\ 4x + y &= 0 \end{aligned} \Rightarrow \begin{aligned} x + 4x + 8x^2 &= 0 \\ y &= -4x \end{aligned} \Rightarrow \begin{aligned} x(8x + 5) &= 0 \\ y &= -4x \end{aligned} \Rightarrow \begin{aligned} x &= 0 \vee x = -\frac{5}{8} \\ y &= 0 \vee y = \frac{5}{2} \end{aligned}$$

$$\underline{EP_1} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \underline{EP_2} = \begin{pmatrix} -5/8 \\ 5/2 \end{pmatrix}$$

JACOBIAN

$$J = \begin{bmatrix} 1-2y & -1-2x \\ 4 & 1 \end{bmatrix}$$

CLASSIFICATION

$$J(0,0) = \begin{bmatrix} 1 & -1 \\ 4 & 1 \end{bmatrix} \Rightarrow \begin{aligned} \tau &= 2 \\ \Delta &= 5 \\ \tau^2 - 4\Delta &= 4 - 20 = -16 \end{aligned} \Rightarrow \begin{aligned} &\text{(check the } \Delta - \tau \text{ diagram)} \\ &(0,0) \text{ unstable spiral} \end{aligned}$$

$$J(-5/8, 5/2) = \begin{bmatrix} -4 & 3/4 \\ 4 & 1 \end{bmatrix} \Rightarrow \begin{aligned} \tau &= -3 \\ \Delta &= -5 \end{aligned} \Rightarrow (-5/8, 5/2) \text{ saddle point}$$

ISOCINES

$$K = \frac{4x + y}{x - y - 2xy}$$

(HORIZONTAL ARROWS)

$$K=0 \Rightarrow 4x + y = 0 \Rightarrow y = -4x$$

(VERTICAL ARROWS)

$$K=\infty \Rightarrow x - y - 2xy = 0 \Rightarrow y = \frac{x}{2x+1}$$

(45° ARROWS)

$$K=1 \Rightarrow x - y - 2xy = 4x + y \Rightarrow y = \frac{-3x}{2+2x}$$

(-45° ARROWS)

$$K=-1 \Rightarrow -x + y + 2xy = 4x + y \Rightarrow x(2y-5)=0$$

$$x=0 \vee y = \frac{5}{2}$$

DIRECTION

$$\dot{x} = x - y - 2xy = 5x + 8x^2 > 0$$

$$\text{ARROWS POINT RIGHT IF } x < -\frac{5}{8} \vee x > 0$$

DIRECTION

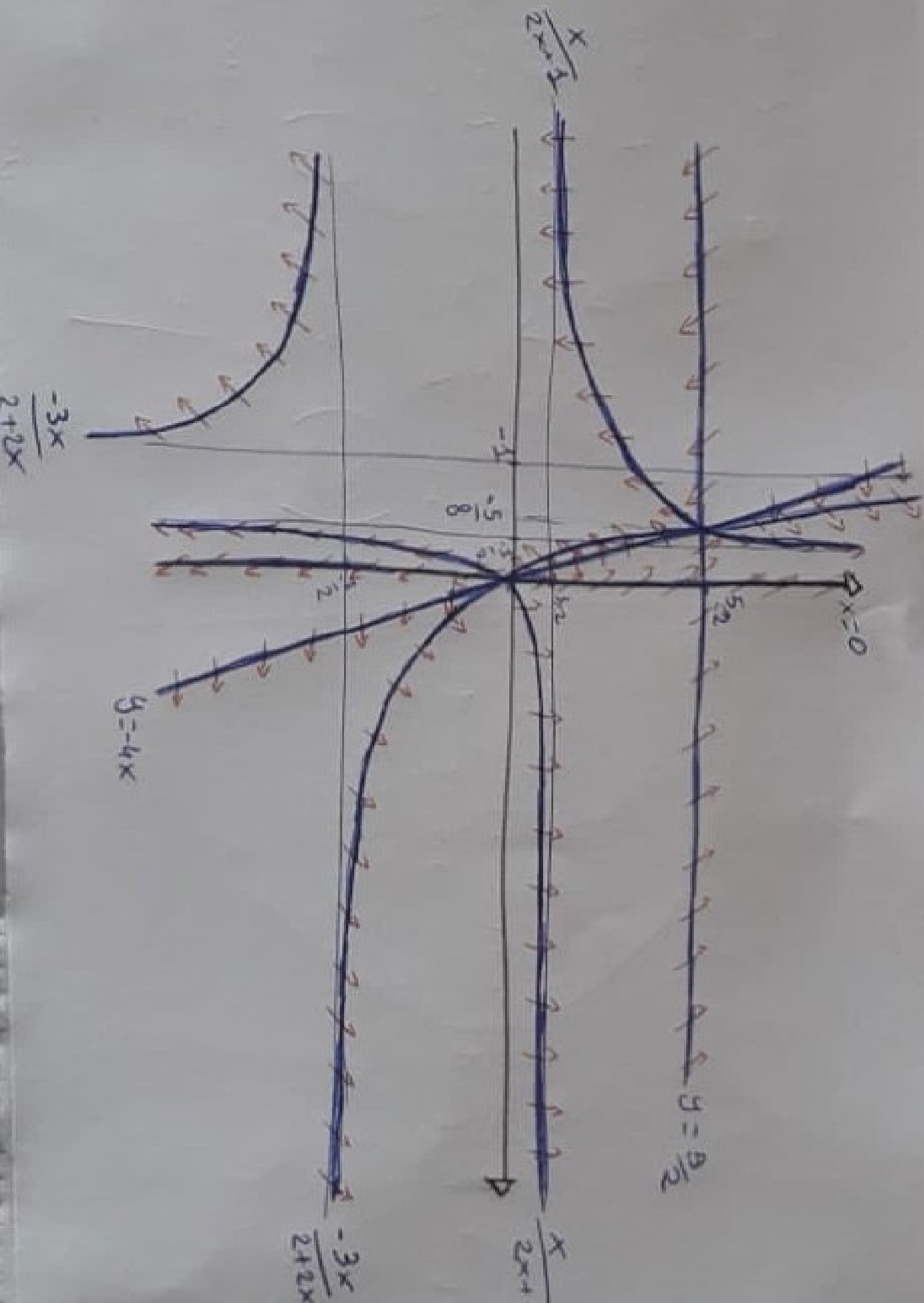
$$\dot{y} > 0 \Rightarrow y > 4x \text{ ARROWS POINT UP}$$

DIRECTION

$$y > 4x \text{ ARROWS POINT } \nearrow$$

DIRECTION

$$y > 4x \text{ ARROW POINT } \nearrow$$



253 Q3

$$\frac{dx}{dt} = y \quad (X(x,y))$$

$$\frac{dy}{dt} = 2 - x - 3x^2 \quad (Y(x,y))$$

HAMILTONIAN

If $\frac{\partial X}{\partial x} + \frac{\partial Y}{\partial y} = 0$ the system is Hamiltonian

$$\frac{\partial X}{\partial x} = 0, \quad \frac{\partial Y}{\partial y} = 0 \Rightarrow \boxed{\frac{\partial X}{\partial x} + \frac{\partial Y}{\partial y} = 0} \quad \checkmark$$

HAMILTONIAN FUNCTION

$$H(x,y) : \quad \frac{\partial H}{\partial x} = -\frac{\partial y}{\partial t} \quad \wedge \quad \frac{\partial H}{\partial y} = \frac{\partial x}{\partial t}$$

$$\frac{\partial H}{\partial x} = -Y(x,y) \quad \frac{\partial H}{\partial x} = 3x^2 + x - 2$$

$$H(x,y) = x^3 + \frac{x^2}{2} - 2x + g(y)$$

$$\frac{\partial H}{\partial y} = +X(x,y) \quad \frac{\partial H}{\partial y} = y \quad H(x,y) = \frac{y^2}{2} + f(x)$$

Thus $H(x,y) = x^3 + \frac{1}{2}(x^2 + y^2) - 2x + C$, let $C=0$

$$\boxed{H(x,y) = x^3 + \frac{1}{2}(x^2 + y^2) - 2x}$$

CLASSIFICATION

$$\frac{\partial H}{\partial x} = 0 \Rightarrow 3x^2 + x - 2 = 0 \Rightarrow \begin{matrix} x = -1 \\ x = \frac{2}{3} \end{matrix} \Rightarrow \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \begin{pmatrix} \frac{2}{3} \\ 0 \end{pmatrix}$$

$$\frac{\partial H}{\partial y} = 0 \Rightarrow y = 0$$

$$\Delta = \frac{\partial^2 H}{\partial x^2} \frac{\partial^2 H}{\partial y^2} - \frac{\partial^2 H}{\partial x \partial y} = (6x+1) \times 1 - 0 = 6x+1$$

$$\Delta(-1,0) = -5 \Rightarrow \text{saddle point}$$

$$\Delta\left(\frac{2}{3},0\right) = +5 \Rightarrow \text{centre}$$

$$H_{xx} \Big|_{\left(\frac{2}{3},0\right)} = 6x+1 \Big|_{\left(\frac{2}{3},0\right)} = 5 > 0 \quad \text{local minimum}$$

DOUBLE CHECK

$$J = \begin{bmatrix} 0 & 1 \\ -1-6x & 0 \end{bmatrix}$$

$$x=0 \\ \Delta = 1+6x$$

C.P.

$$y=0$$

$$2-x-3x^2=0$$

$$LP_1 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \quad LP_2 = \begin{pmatrix} \frac{2}{3} \\ 0 \end{pmatrix}$$

CLASSIFICATION

$$LP_1 : \begin{matrix} x=0 \\ \Delta = -5 \end{matrix} \Rightarrow \text{saddle point}$$

$$LP_2 : \begin{matrix} x=0 \\ \Delta = 5 \end{matrix} \Rightarrow \text{centre}$$

PHASE PORTRAIT

Let us study the function $H(x,y) = x^3 + \frac{1}{2}(x^2 + y^2) - 2x$

$$H\left(\frac{2}{3}, 0\right) = \frac{8}{27} + \frac{2}{9} - \frac{4}{3} - \frac{4}{3} = \frac{8+6-36}{27} = -\frac{22}{27}$$

$$H(-1, 0) = -1 + \frac{1}{2} + 2 = \frac{3}{2}$$

ASYMPTOTIC ANALYSIS

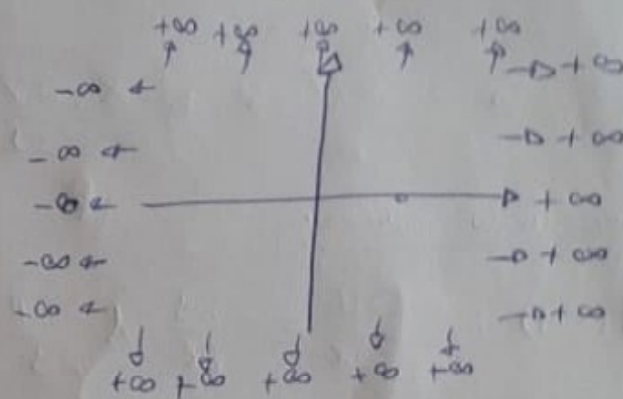
$$\lim_{x \rightarrow +\infty} H = +\infty \quad \text{for fixed } y$$

$$\lim_{x \rightarrow -\infty} H = -\infty \quad \text{" "}$$

$$\lim_{y \rightarrow +\infty} H = +\infty \quad \text{" "}$$

$$\lim_{y \rightarrow -\infty} H = +\infty \quad \text{" "}$$

H assumes all values between $(-\infty, +\infty)$



CONSTANT HAMILTONIAN

where is $H(x,y) = c$ for c constant?

- At the centre $x = \frac{2}{3}, y = 0 \Rightarrow H = -\frac{22}{27}$

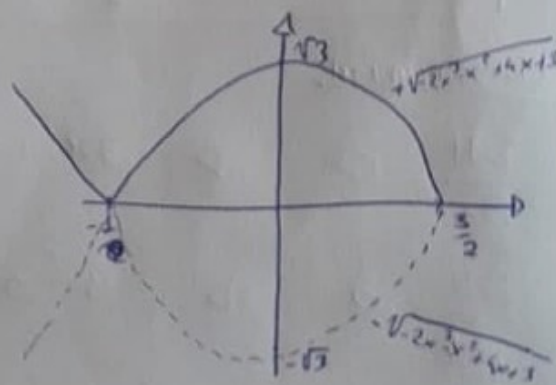
- There are two curves that intersect at the saddle point $x = -1, y = 0, H = \frac{3}{2}$

$$x^3 + \frac{1}{2}(x^2 + y^2) - 2x = \frac{3}{2}$$

$$\frac{y^2}{2} = -x^3 - \frac{1}{2}x^2 + 2x + \frac{3}{2}$$

$$y^2 = -2x^3 - x^2 + 4x + 3$$

$$y = \pm \sqrt{-2x^3 - x^2 + 4x + 3}$$

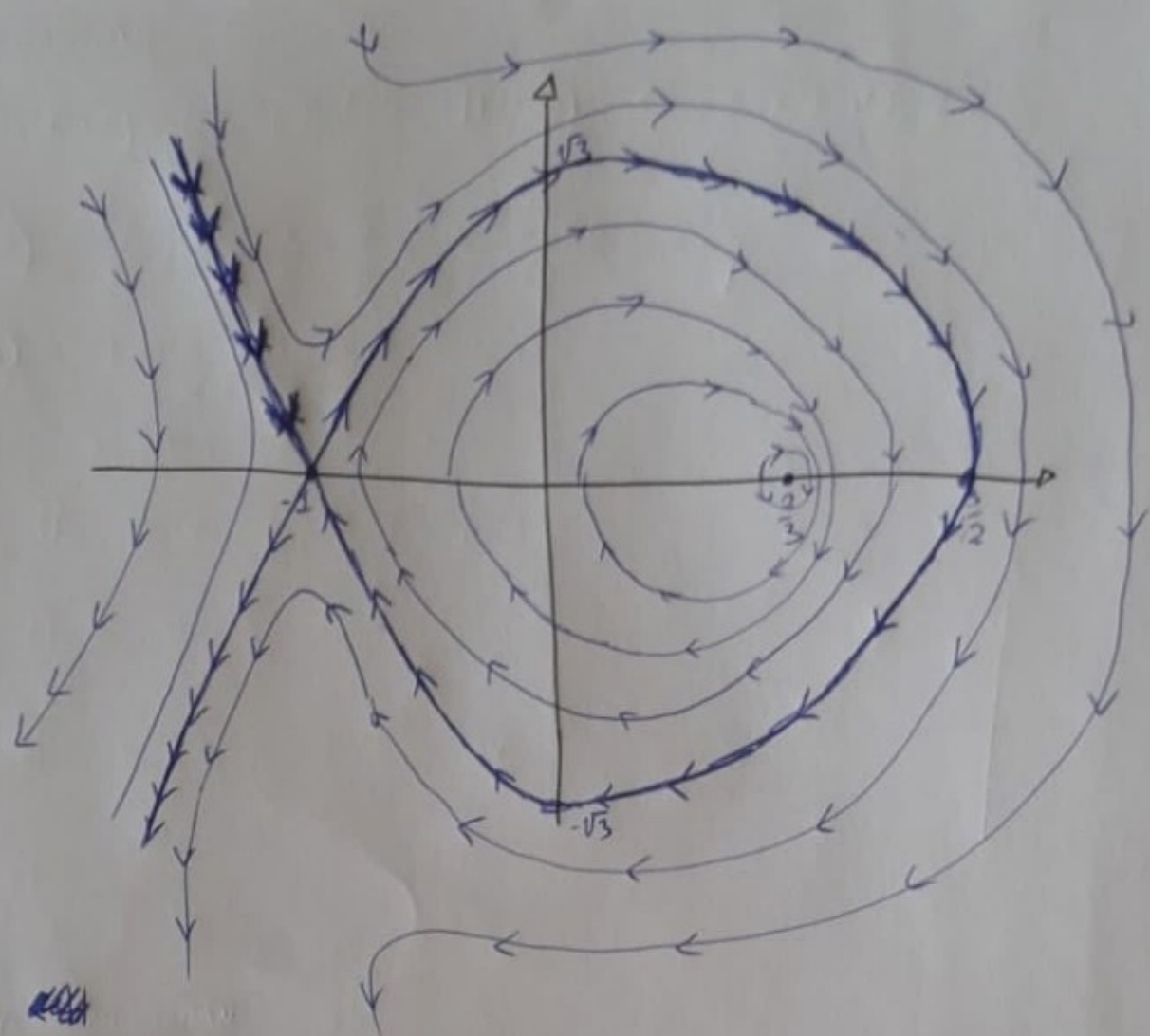


the curves cut the vertical axis at $x=0 \Rightarrow y = \pm\sqrt{3}$

the curves cut the horizontal axis at $y=0 \Rightarrow (x+1)^2(x-\frac{3}{2})=0$

$x = -1$ (we already knew this, since they cross in the saddle point)

$$x = \frac{3}{2}$$



To find the arrows direction consider the point $(\frac{3}{2}, 0)$

$$\dot{x} = 0$$

$$\dot{y} = 2 - \frac{3}{2} - 3\left(\frac{y}{4}\right) = -6.25 \quad \text{ARROW POINTS DOWN}$$

Once we know the direction of the arrow in $(\frac{3}{2}, 0)$ we can figure out all the other arrows by just following the flow.